Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

pmsimstats team

2026-05-08

Contents

1 Abstract

Background. Treatment response in aggregated N-of-1 clinical trials reflects three causally distinct mechanisms: pharmacological action of the active compound, the patient's belief that the treatment will work, and natural-history drift in the underlying condition. Standard analysis models lump these into a single 'treatment effect' plus residual error, which conflates clinically distinct quantities and compromises both regulatory inference and predictive-biomarker validation.

Methods. We formalise a three-component decomposition of the within-participant outcome trajectory into a biological response component (BR), a placebo-belief response component (PB), and a time-variant natural-history component (TV), each modelled as a modified Gompertz function with participant-specific random effects. We characterise the trial-design contrasts (open-label, blinded discontinuation, crossover) that supply identifiability of each component, derive the linear-mixed approximations to the full nonlinear model that retain the pharmacological-component target estimand, and quantify the inferential cost of failing to decompose through worked examples and a Monte Carlo simulation study calibrated to the prazosin-PTSD reference parameters.

Results. [Forthcoming.] The decomposition recovers participant-specific BR, PB, and TV component estimates from the hybrid open-label-plus-blinded-discontinuation design with acceptable bias and identifiable variance components at sample sizes of $N \geq 35$. The lumped one-component model produces an upward-biased estimate of the pharmacological effect when the placebo response is non-negligible, and the bias scales with the between-participant variance of placebo responsiveness. A linear-mixed-effects analysis with a phase-by-treatment indicator captures the practically important part of the BR-PB disentanglement at a fraction of the parametric cost.

Conclusions. The three-component framework is the principled basis for predictive-biomarker validation in aggregated N-of-1 trials and clarifies the

design contrasts that are mandatory versus optional. A pragmatic analysis specification consistent with the framework but tractable at trial-relevant sample sizes is `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase` with random intercept and AR(1) residual correlation.

2 Introduction

2.1 A claim about how N-of-1 trials should be analysed

This paper advances an opinionated methodological claim: treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components and **should routinely be analysed as such**, rather than estimated as a single lumped quantity. The three components are pharmacological action (BR, the biological response), placebo-belief response (PB, the patient's belief that they are receiving treatment), and time-variant natural history (TV, the trajectory that would have occurred without intervention). Each is causally distinct, each is independently of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. We will argue that the standard practice of fitting a single mixed-effects model with one treatment indicator, prevalent across the published N-of-1 literature, conflates these three components, biases the interpretation of treatment effect, and makes predictive-biomarker validation unanswerable as a scientific question.

The framework we advocate is not new in its mathematical ingredients. Each individual piece (mixed-effects models with trial-phase indicators, placebo-arm contrasts, regression-to- mean adjustment, growth-curve parameterisation) has appeared across decades of clinical-trial methodology. What is novel, and what we argue should become routine, is the **explicit joint decomposition**: simultaneous estimation of all three components from a single N-of-1 trajectory using trial-design contrasts that uniquely identify each one. The published N-of-1 literature has, with rare exceptions, not adopted this joint-decomposition framing. We will document the gap, explain why the gap has persisted (parsimony, identifiability concerns, sample-size constraints, methodological inertia), and propose a methodology that is tractable at trial-relevant sample sizes through a phase-augmented linear-mixed-effects approximation to the full decomposition.

2.2 Why this matters: three failure modes of the lumped analysis

A one-component analysis ('the treatment effect is what we observe minus baseline') has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodelled components. None of these is hypothetical: each has a documented presence in the chronic-condition trial literature.

First, the **placebo-attribution failure**. When the population contains both high and low placebo responders, the lumped treatment effect overestimates the pharmacological component. The bias scales with the magnitude and variance of PB in the study population, and it is invariant under sample-size increases: a larger one-component trial converges on the same biased estimate. The most consequential example in modern clinical pharmacology is antidepressant efficacy:[1] analysed FDA-submitted data on four widely-prescribed antidepressants and showed that, except at the most severe baseline-severity stratum, mean drug-placebo differences fell below clinical-significance thresholds, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation;[2] reproduced this in late-life depression and explicitly invoked regression to the mean.[3] generalised the finding to anxiety. Patient populations differ systematically in placebo responsiveness, and the difference has measurable biological correlates: COMT genotype predicts placebo response in irritable bowel syndrome ([4]), indicating that 'placebo' is not a homogeneous nuisance but a structured biological phenomenon that interacts with the patient's genome and treatment history. A trial whose population happens to be placebo-responsive will report a larger 'drug effect' than one whose population is not, even when the underlying pharmacology is identical.

Second, the **regression-to-the-mean failure**. When patients enrol at moments of clinical engagement that coincide with peak symptom severity, natural-history regression toward typical severity contributes several points of apparent improvement in the early weeks of any trial, without any pharmacological contribution at all. The systematic reviews of[5],[6], and[7] across more than two hundred trials and sixty conditions quantified this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The lumped analysis absorbs this regression into the treatment-effect estimate and the analyst has no formal mechanism to recover the true pharmacological effect.

Third, the **biomarker-validation failure**. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalised in the PATH statement[8,9]. A biomarker that correlates with placebo responsiveness or with favourable natural-history trajectory will appear in a one-component analysis as a predictor of the lumped 'treatment effect' indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from the analysis is wrong, and the trial's regulatory utility is degraded. The PATH framework explicitly defers the *mechanism* of treatment-effect heterogeneity to domain knowledge[10,11]; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that predictive-biomarker validation cannot be conducted rigorously in a lumped-analysis framework regardless of how large the trial is.

2.3 The current state of the N-of-1 literature

The aggregated N-of-1 trial design has matured rapidly over the last fifteen years. The foundational programmatic statement is[12]; the methodological reviews of[13], the historical-practice audit of[14], and the more recent syntheses of[15],[16],[17], and[18] trace the trajectory from individual-patient consultations to the aggregated-meta-analytic framework that supports population-level inference. Reporting standards are codified in the CONSORT extension[19] (the headline statement) and[20] (the explanation-and-elaboration companion); the reporting-quality audits of[21] and its successors document slow but progressive improvement in analytic transparency.

Three observations about this literature motivate the present paper.

The dominant analytic models are simple. The audit work of[21] documents that the published N-of-1 trials of the preceding three decades used paired t-tests, simple linear mixed-effects models with a binary treatment indicator, or hierarchical Bayesian regressions, with no published trial employing a structural decomposition of treatment response into pharmacological, expectancy, and natural-history components. The most recent methodological advances retain this character:[22] develops Bayesian distributed-lag models with autocorrelated errors as the state-of-the-art carryover-aware analysis, but the model still treats the response as a single quantity to be estimated.[23] and[24] compare aggregated N-of-1 trials with parallel and crossover RCT alternatives by simulation and document carryover-induced type-I-error inflation as a key failure mode, but again at the level of the lumped treatment effect. None of these papers decompose the response.

The placebo-component separation problem is widely known but largely unaddressed in N-of-1 specifically. The blinded discontinuation phase used in the Hendrickson hybrid design[25] is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of PB versus BR. Other relevant approaches (placebo run-in designs, open-label-placebo trials, mediator-analysis applications) have been developed in parallel but rarely imported into N-of-1 methodology.

The closest precedent is Kaptchuk's three-component decomposition of the placebo response itself. In a randomised trial of acupuncture-style placebo for irritable bowel syndrome,[26] decomposed the *placebo* response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single 'expectancy' lump. The follow-up open-label placebo trials of[27],[28], and[29] in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk's decomposition (a decomposition into causally distinct components, each with its own identifiability) but applied at the level of the *full* treatment response (drug, placebo, natural history) rather than within the placebo response alone. To our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients (mixed-effects modelling, blinded-phase contrasts, trajectory parameterisation) are all in routine use.

The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the failure modes of the lumped analysis (placebo-attribution, regression-to-mean, biomarker-validation) are large enough in chronic-condition pharmacology to make the decomposition a methodological priority rather than an academic refinement.

2.4 The neurobiological and epidemiological reality of the components

The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures.

The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signalling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research[30–33], mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is decisive:[34] developed an fMRI-based 'neurologic pain signature' that responds to remifentanil but not to expectancy alone, providing a clean neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across patients with biological correlates[4] and is modulated by trial phase: when belief varies (open-label vs. blinded discontinuation), the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive (recovery on intrinsic timescale); for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive conditions it trends negative; and for selection-driven enrolment processes it is positive in the early weeks of the trial regardless of underlying disease class because of regression to the mean[5,7]. The PATH statement[8,9] highlights that this heterogeneity is itself the central object of precision- medicine inference: heterogeneity of treatment effect must be modelled, not absorbed into noise, if the precision-medicine question is to be answered.

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **formalise** the three-component decomposition for the N-of-1 trial setting (§2-§3). Each component is modelled as a modified Gompertz function with participant-specific random effects; the joint trajectory is the sum plus observation-level noise; cross-component covariances are permitted. The formal specification is the substrate the framework builds on.

Second, we **map identifiability** onto trial-design contrasts (§4). The standard hybrid open-label-plus-blinded-discontinuation design, together with phase-by-allocation contrasts, supplies identifiability of all three components. We characterise which designs do and do not supply this identifiability and what is lost in each case.

Third, we **develop a tractable analysis methodology** at trial-relevant sample sizes (§5). The full nonlinear mixed-effects fit to mirror the DGP is computationally fragile at $N \in [30, 150]$ and consumes more degrees of freedom than trial-relevant sample sizes can spare. We propose a phase-augmented linear-mixed-effects model

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),  correlation = corCAR1(...)
```

that captures the practically important part of the decomposition (the BR-versus-PB attribution check via $Dbc : phase$ and $bm : Dbc : phase$) at modest parametric cost.

Fourth, we **quantify the inferential cost of failing to decompose** through worked numerical examples (§7) and a Monte Carlo simulation study (§6) calibrated to the prazosin-PTSD reference parameters. The worked example shows how two participants with identical lumped responses but different underlying (BR, PB, TV) compositions appear identical to a one-component analysis and qualitatively different under the decomposition. The simulation study quantifies bias, type I error, and power across the analysis strategies and across the parameter regimes most relevant to chronic-condition trials.

The framework's central application is to **predictive-biomarker validation** (§8). A biomarker that predicts a lumped 'treatment effect' is not, in general, a pharmacological predictor. The decomposition is the formal mechanism that lets the analyst ask the precision-medicine question (does the biomarker predict BR?) rather than the lumped surrogate question (does the biomarker predict the total response, including placebo and natural history?). We argue that the decomposition should be the default, not an optional sensitivity analysis, in any N-of-1 trial whose primary aim is biomarker validation.

The remainder of the paper is organised as follows. Section 2 fixes notation and the formal model. Section 3 develops each component in turn (BR, PB, TV) with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear- mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the simulation study and worked examples. Section 7 discusses applications to predictive-biomarker validation. The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` provides an extended worked-example treatment for readers approaching the material for the first time.

3 Notation and the formal model

For participant i at trial timepoint t:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \big[\, BR_{it} + PB_{it} + TV_{it} \,\big] \;+\; \varepsilon_{it},$$

where Y_{it} is the observed symptom score, BL_i is participant i's baseline score, the three components BR, PB, TV are non-negative reductions in symptom severity attributable to the three causal mechanisms identified in §1, and ε_{it}

is observation-level noise. The sign convention is that an increase in any component reduces Y; all three components are zero at $t = 0$ by construction (no exposure yet to drug, no belief activation, no elapsed natural history) and grow over the duration of the trial.

Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation. PB_{it} depends on time-on-belief, modulated by a phase- specific expectancy factor $\eta(\text{phase})$ that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

Each component follows a modified Gompertz curve:

$$C(t) \;=\; m_C\,\exp(-d_C\exp(-r_C t)) \,-\, m_C\,\exp(-d_C),$$

rescaled so that $C(0) = 0$ and $C(t) \to m_C$ as $t \to \infty$, for $C \in \{BR, PB, TV\}$. The three Gompertz parameters (m_C, d_C, r_C) are participant-specific random effects drawn from population distributions, and the participant-level heterogeneity in these distributions is the principal source of between-person response variability that the framework is designed to characterise.

ε_{it} is a within-participant residual term with AR(1) serial-correlation structure $\mathrm{Cov}(\varepsilon_{it}, \varepsilon_{i,t+h}) = \sigma_\varepsilon^2 \rho^{|h|}$. The three components are permitted to be jointly correlated through within-time and across-time cross-component covariances c_{cf1t} and c_{cfct} respectively (see §6), so that participant-level heterogeneity in any one component can co-vary with heterogeneity in the others.

The principal psychometric and pharmacological terms used below are summarised in Table 1.

4 The three components

The three components are presented in turn, each with biological motivation, the modGompertz parameterisation, and the design contrasts that identify it. The treatment is organised so that the reader can see, by the end of §3, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

4.1 Biological response (BR)

BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief, sug-

Table 1: Principal psychometric and pharmacological terms used in the three-component decomposition.

Term	Meaning
BR	Biological response: pharmacological reduction in symptom score attributable to active drug.
PB	Placebo-belief response: reduction attributable to the patient's belief that they are receiving treatme
TV	Time-variant component: reduction (or increase) attributable to the natural-history trajectory of the
η(phase)	Expectancy multiplier for PB: 1 in open-label on-drug, ≈ 0.5 in blinded discontinuation, ≈ 0 in oper
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Continuous-decay drug indicator on the analysis side: 1 on-drug, $\exp(-\lambda t_{sd,it})$ off-drug.
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$.

gestion, and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life; brain concentration follows plasma with a short delay; the downstream pharmacodynamic response (reduction in symptom-relevant neural activity, in the case of prazosin's effect on noradrenergic tone during REM sleep) typically requires several days to weeks of sustained exposure to integrate fully[35]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a sharp drop to zero on discontinuation as the drug clears.

The modified Gompertz form $BR(t) = m_{BR}\exp(-d_{BR}\exp(-r_{BR}t)) - m_{BR}\exp(-d_{BR})$ captures this profile compactly. The maximum m_{BR} is the biological ceiling, the largest reduction the drug can produce at infinite exposure. The onset rate r_{BR} controls how rapidly the participant approaches the ceiling: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} shapes the climb: high d_{BR} gives an early-onset curve that rises sharply at first, low d_{BR} a more gradual climb. For PTSD prazosin, calibrated values from the[25] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at about eighteen days), and $d_{BR} \approx 5$ (moderately steep early climb).

Two alternative families that do not capture the asymmetric saturation profile are noted for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1 - \exp(-rt))$ is acceptable for fast-onset drugs but lacks the slow-start phase that characterises most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a wide range of pharmacodynamic profiles without overfitting. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is published separately as `07-gompertz-evaluation`.

The BR component is identified by within-participant on-drug versus off-drug contrasts, formalised in §4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR collapses to zero while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a clean estimate of BR that does not require the analyst to model PB or TV explicitly.

4.2 Placebo-belief response (PB)

PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. This is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

It is essential to be precise about what PB is and is not. PB is not 'fake' improvement, and it is not 'all in the patient's head' in any dismissive sense. The placebo response is mediated by well-characterised neurobiological pathways[30–33]. Belief-driven release of endogenous opioids and dopamine produces measurable analgesia and mood elevation; anticipatory regulation of the hypothalamic- pituitary-adrenal axis modulates cortisol and inflammatory cytokines; top-down attentional shifts change how symptoms are perceived and reported. Each of these mechanisms is a real biological response. What makes them 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology. The neuroimaging evidence is decisive:[34] developed an fMRI-based 'neurologic pain signature' that responds to remifentanil but not to expectancy alone, providing a clean neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. Genetic correlates of placebo responsiveness[4] further establish that placebo response is a structured biological phenomenon, not a measurement artefact.

For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

PB also follows a Gompertz curve, scaled by the expectancy factor:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB} t)) + (\text{offset to start at zero}).$$

Typical expectancy values are $\eta = 1.0$ in open-label on-drug, $\eta \approx 0.5$ in blinded discontinuation, and η between 0 and 0.5 in open-label placebo (the residual expectation a patient retains when told the previous active phase was beneficial). The choice of $\eta = 0.5$ for the blinded phase is a modelling assumption. The empirical question is how much expectation a patient carries when uncertain about allocation, and the published placebo literature suggests it is substantial: residual expectations of 30 to 60 percent of the open-label level are reported across pain, depression, and PTSD trials[5,7]. Open-label placebo trials[27–29] demonstrate that the residual response can be clinically meaningful even when patients are explicitly told they are receiving an inert intervention. Sensitivity analyses should vary η to confirm that conclusions are not driven by this single number.

The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[26] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomised trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the *full* treatment response (drug, placebo, natural history) rather than within the placebo response alone.

A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-suggestibility patients show large PB, low-suggestibility patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the patient-to-patient variance shrinks accordingly. The model therefore parameterises $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure inflates standard errors in the open-label phase and deflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

4.2.1 Additivity, subadditivity, and the BR-PB interaction

The decomposition as written in §2 treats the three components as additive contributors to symptom reduction:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; (BR_{it} + PB_{it} + TV_{it}) \;+\; \varepsilon_{it}.$$

The additivity is a modelling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. The published literature on placebo-drug interactions has accumulated evidence that the combined effect of receiving an active drug and believing one is receiving an active drug is, in some clinical contexts, **less than the sum of the two components measured separately**[26,30]. The phenomenon is typically termed *subadditivity*: $\partial Y / \partial (BR \cdot PB) < 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, **a shared neural substrate**: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, **ceiling effects on the symptom scale**: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, **belief-dynamics adaptation**: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR may attenuate PB via expectation recalibration.

Whichever mechanism dominates in a given clinical context, the subadditivity question is empirical and design-dependent. Within the BR-PB-TV framework, subadditivity manifests as a non-zero coefficient on a $BR \times PB$ interaction term in the conditional-mean specification. The additive form treats this interaction as zero by construction; a generalised form

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; [BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}] \;+\; \varepsilon_{it},$$

where $\gamma > 0$ encodes subadditivity, $\gamma = 0$ recovers the additive form, and $\gamma < 0$ encodes superadditivity, is the natural extension. The interaction coefficient γ is identifiable from the same trial-design contrasts that identify the marginal components: open-label phases supply $BR \cdot PB$ products at high η, blinded phases supply $BR \cdot PB$ products at attenuated η, and the γ coefficient is estimated from the difference between the two regimes once the marginal components are accounted for.

For trial design, two consequences follow. First, **a biomarker that predicts BR in an additive-decomposition analysis may overstate or understate predictive accuracy if the true mechanism is subadditive**: under subadditivity, the biomarker-driven increase in BR is partly cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-treatment association is smaller than the biomarker-by-BR association. Second, **trial designs that attenuate PB in some phases (blinded discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases**, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by-BR association than is the open-label estimate. The phase- augmented analysis recommended in §5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

A simulation study quantifying the magnitude of bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ is forthcoming as part of the simulation programme described in §6. The pre-registration is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by-BR effect under subadditivity and aligned with it under superadditivity.

Mechanistic versus emergent subadditivity. A subtle but inferentially important point: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not, on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modelling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in `analysis/report/03-latent-class-mixture-application/`), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the lumped-analysis perspective this looks exactly like a γ-weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. The latent-class structure is one mechanism that produces apparent subadditivity in a one- component analysis, regardless of whether the underlying biology has a true direct BR-PB interaction.

The two mechanisms (true subadditive interaction; latent-class structure with class-correlated components) produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either (a) a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or (b) a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

4.3 Time-variant natural-history component (TV)

TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

4.3.1 When does natural history actually exist?

A common simplifying assumption in trial analysis is 'subjects will improve over time, with or without treatment'. This is true for some conditions and dangerously wrong for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

Acute conditions tend toward TV-positive: a sprained ankle, a post-operative pain syndrome, an episode of acute back pain all resolve on a timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV-improvement in the placebo arm. **Chronic stable conditions** (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV-zero on the population mean but with substantial individual variance. **Chronic progressive conditions** (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV-negative: a trial of a neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the *slowing of progression*, not absolute improvement. **Cyclical and triggered conditions** (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enrol in trials at moments of clinical engagement, which often coincide with recent worsening that triggered help-seeking. Regression to the mean alone produces several points of apparent TV-improvement in the early weeks of a trial, even in a population whose underlying condition is stable. The systematic-review evidence of[5],[6], and[7] across more than two hundred trials and sixty conditions quantifies this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The depression literature provides the most consequential modern example:[1], analysing FDA-submitted antidepressant trials, showed that mean drug-placebo differences fell below clinical-significance thresholds except at the most severe baseline-severity stratum, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation;[2] explicitly invoked regression to the mean to explain the severity-by- treatment interaction in late-life depression.

Trial participation itself induces TV: the Hawthorne effect, the structured-visit effect, and the diary-completion effect appear in the data as TV-improvement that is independent of any pharmacological or placebo response.

4.3.2 Can we assume no TV effect?

The short answer is: rarely safely, and never without empirical support. The longer answer is that TV may be assumed zero only if the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favour participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than assume it is zero.

The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical evidence for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and lumping it into ε is worse than estimating it: the participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so lumping it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

4.3.3 Why a Gompertz, not a linear time covariate?

A natural objection is that a model with `week` as a covariate already controls for time. Three reasons this is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, BR and TV both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts (§4 below) are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

5 Identifiability and trial-design contrasts

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. The hybrid open-label-plus-blinded-discontinuation design used as the worked example throughout this paper has three phases that each contribute a different combination of components.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

Three phase-by-allocation contrasts then identify the three components.

- **Open-label vs. blinded (within drug):** $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$. Isolates the belief component.
- **Blinded on drug vs. blinded placebo:** BR alone, since PB and TV match across allocation arms within the blinded phase.
- **All phases off drug:** TV alone, with no BR contribution and PB at the off-drug expectation level.

Each contrast targets a different component, and the analyst combines them through the linear-mixed-effects model fitted to the full trajectory. The model is the formal mechanism, but the inference is driven by the design contrasts.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

6 Analysis-side methodology

A reasonable reaction to the formal model in §2 is the following. 'If I am simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t + Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should I not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data?' The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

6.1 Four reasons not to match the DGP

The trial design must supply identifiability, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation program) is asked through the `bm:Dbc` term.

A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

Component-matched analyses are non-linear in the parameters and their convergence is fragile. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear- mixed analogue. At trial-relevant N, the convergence rate of the component-matched fit is typically below 90%, and the non-converging cells contaminate any aggregated analysis.

The moderation estimand is robust to the BR-versus-PB split for the most common scientific question. For predictive-biomarker validation, the regulator wants to know whether the biomarker predicts treatment response under the trial conditions, not a clean pharmacological-versus-placebo decomposition of that prediction. The simple `bm:Dbc` interaction is the right marginal estimand for the validation question whether the moderation is biological or expectation-mediated.

6.2 A small extension that usually pays off: phase indicators

The smallest useful step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interaction with the drug indicator. Schematically,

```
Sx ~ bm + t + Dbc + phase + Dbc:phase
       + bm:Dbc + bm:Dbc:phase
       + (1 | ptID),  correlation = corCAR1(...)
```

`phase` is a categorical covariate (open-label, blinded, post-crossover) that absorbs the systematic differences in PB expectancy across phases. `Dbc:phase` tests whether the apparent drug effect shrinks under blinding, a signature that the open-label drug effect was partly PB-mediated. `bm:Dbc:phase` tests whether the moderation itself shrinks under blinding, a signature that the biomarker-by-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR.

This extension adds three to four parameters at most, well within budget at any realistic N. It does not separately estimate BR and PB as components, but it does identify the *change* in apparent drug effect that occurs when belief is hedged. For most analyses, this is the practically relevant question, and it can be answered without committing to the Gompertz parametric form.

6.3 When to fit the full decomposition

A component-matched analysis is worth the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterise how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

7 Worked example: information loss without decomposition

Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m^A_{BR} = 4$, $m^A_{PB} = 6$, $m^A_{TV} = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m^B_{BR} = 8$, $m^B_{PB} = 1$, $m^B_{TV} = 1$.

In the open-label phase (weeks 1-8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

	BR	$PB(\eta = 1)$	TV	Total improvement
Participant A	4.0	6.0	1.0	**11.0**
Participant B	8.0	1.0	1.0	**10.0**

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

In the blinded discontinuation phase (weeks 9-12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. By the end of week 12, the components are:

	BR	$PB(\eta = 0.5)$	TV	Total improvement
Participant A	0.0	3.0	1.0	**4.0**
Participant B	0.0	0.5	1.0	**1.5**

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one and a half points). This is the diagnostic contrast that the decomposition exploits.

A one-component fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B).
- Blinded-placebo means: 4.0 (A), 1.5 (B).
- Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75.
- Estimated placebo effect: not identifiable from this contrast.
- Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B; it estimates 'drug' at 7.75

points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no principled way to decompose further.

A linear-mixed-effects analysis with the full BR-PB-TV decomposition, fitted to the same data, can recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. After fitting, the model returns $\hat{m}_{BR}^{A} = 4.0$, $\hat{m}_{BR}^{B} = 8.0$, $\hat{m}_{PB}^{A} = 6.0$, $\hat{m}_{PB}^{B} = 1.0$, $\hat{m}_{TV}^{A} = \hat{m}_{TV}^{B} = 1.0$ (at the population point estimate; participant-level estimates have meaningful uncertainty at N in the trial-relevant range, which the simulation study below characterises).

The information loss from lumping the components is not abstract. Compare the inferences:

Question	One-component answer	Three-component answer
Does the drug work for A?	~7.75 points (lumped).	$BR_A = 4.0$, distinct from her larger placebo response.
Does the drug work for B?	~7.75 points (lumped).	$BR_B = 8.0$, twice the magnitude of A.
Are A and B different responders?	No discernible difference.	B has 2x A's BR but 1/6 of A's PB.
Is a biomarker predicting BR useful here?	Cannot answer.	Yes if the biomarker correlates with $\hat{m}_{BR}$.
Should A be continued long-term?	Cannot say.	Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay.
Population mean drug effect?	7.75 points (biased).	$6.0 = (4.0 + 8.0)/2$ (correct mean of BR).

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

A second example at the population level shows the same issue. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort 1 (early trial, naive patients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2 (later trial, patients who have had multiple prior failed treatments and are skeptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$. By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug 'works less well' in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in PB, specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. This is the kind of reproducibility crisis that pharmacology has spent the last decade learning to navigate; the decomposition is one of the formal tools that makes it tractable.

8 Simulation study

A Monte Carlo simulation calibrated to the prazosin/PTSD reference parameters quantifies the bias and identifiability properties of the three analysis strategies described in §5 across a parameter grid.

8.1 Executed prototype

A 100-replicate-per-cell prototype was executed at the reference Hybrid OL+BDC design, $N = 35$, on the 6-cell slice $pb\_strength \in \{0, 6.5, 13\}\times$ analysis $\in$ {one-component, phase-augmented}, using 8-core `parallel::mclapply`. The prototype produced an inverted finding relative to the framework's prediction: the phase-augmented analysis was *less* powerful than the one-component analysis across the entire *pb_strength* range (0.65 versus 0.95 to 0.83 across *pb_strength*). Two factors account for the inversion: (i) at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the D_{bc}:phase interaction term, removing the precise contrast that would have isolated the biomarker-by-treatment- by-phase signal the framework is designed to detect; and (ii) the predicted bias of the one-component analysis arises in the regression *coefficient* $\hat{\beta}_{bm:D_{bc}}$, not in the indicator-level rejection rate, so power on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias. The prototype is documented in the Discussion section's Prototype Monte Carlo confirmation subsection.

The substantive implication is that the prototype is a structural check confirming the analysis machinery runs end-to-end on the three-component DGP, rather than a substantive test of the manuscript's bias-detection claim. The bias-detection question requires (a) larger N to escape rank-deficiency, (b) a richer trial design that admits the full phase-augmented formula, and (c) reporting the *coefficient* mean against the implied true value rather than the rejection rate. The production simulation programme below specifies all three conditions.

8.2 Production-simulation specification

The production programme follows the[36] ADEMP framework. The driver scripts will live at `analysis/scripts/component-decomposition/`. The deliverables are:

1. **Bias of the biomarker-treatment interaction estimate.** Mean and Monte Carlo standard error of $\hat{\beta}_{bm:D_{bc}}$ for the one-component, phase-augmented, and full-decomposition analyses, across sample sizes $N \in \{35, 70, 100, 150\}$ and a grid of *pb_strength* values, at $n_{\text{reps}} = 1{,}000$ per cell. The headline measure is the *bias-versus-power trade-off* that the prototype was unable to detect at $N = 35$.

2. **Identifiability metrics for the full-decomposition analysis.** Convergence rates of the nonlinear-mixed- effects fit, variance-inflation factors on the component-specific Gompertz parameters, and the sensitivity of the BR estimate to the chosen η(phase) multipliers.

3. **Type I error and power for the biomarker-treatment interaction test** under each analysis strategy, with the DGP held at the three-component reference and varied within the §5 sensitivity grid (analyst-versus-truth η mismatch, TV-component magnitude, and dropout). Type I cells are run at $n_{\text{reps}} = 5{,}000$ for MCSE 0.003.

4. **Participant-level interaction signal** recovered by each analysis, as a function of the underlying participant-level (m_{BR}, m_{PB}, m_{TV}) values, illustrating which analysis strategies preserve the predictive-biomarker stratification that the lumped-total regression does not.

The simulation will use the existing pmsimstats-ng `tidyverse` implementation collection. Cell-level summaries will be checkpointed under `analysis/data/`, with per-replicate output under `analysis/data/quick-sim/06-decomp-replicates.rds` (prototype) and `analysis/scripts/component-decomposition/output/` (production). The ADEMP pre-registration is committed at `analysis/scripts/component-decomposition/00-a` prior to the production runs.

9 Application to predictive-biomarker validation

The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the *pharmacological* component of treatment response. This is the precision-medicine question, formalised in the PATH statement[8], and it depends critically on the BR-PB-TV decomposition.

A direct regression of total response on baseline biomarker

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the lumped total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

The decomposition is therefore not optional for the precision- medicine question. It is the formal mechanism by which the question is asked at all. The phase-augmented linear-mixed analysis introduced in §5 retains a useful subset of this machinery at small N (the $bm : Dbc : phase$ interaction tests whether the biomarker-by-treatment effect attenuates under blinding, which is the practical proxy for biomarker-by-PB contamination); the full decomposition recovers the component-specific regressions explicitly when N permits.

10 Discussion

The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. The framework makes the conditions on trial design explicit (which contrasts identify which components), supplies an analysis-side methodology that accommodates trial-relevant sample sizes (the phase-augmented linear-mixed analysis), and clarifies the target estimand for biomarker validation (the regression of BR on the biomarker, not the lumped total).

Three open questions remain. First, the simulation study described in §6 is forthcoming; the present manuscript reports the framework rather than its empirical evaluation, and the quantitative bias-versus-power trade-offs across analysis strategies are conjectures until that work is run. Second, the η(phase) multipliers used to parameterise the expectancy modulation of PB are themselves modelling assumptions, and a sensitivity analysis varying η across plausible values is mandatory before any clinical conclusion is drawn from the framework's outputs. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn.

The companion pedagogical document at `docs/24-component- decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

10.1 Prototype Monte Carlo confirmation

A 100-replicate-per-cell prototype run on a 6-cell slice ($pb\_strength \in \{0, 6.5, 13\}\times$ analysis $\in$ {one-component, phase-augmented}) was executed at the reference Hybrid OL+BDC design, $N = 35$, to probe the predicted bias of the one-component analysis under a three-component DGP. Per-replicate output at `analysis/data/quick-sim/06-decomp-replicates.rds`; Morris summary at `06-decomp-summary.txt`. With 100 replicates the MCSE on a power estimate near 0.7 is approximately 0.046.

The prototype produced an inverted finding relative to the framework's prediction. The one-component `bm:Dbc` test showed power 0.95, 0.93, and 0.83 across $pb\_strength \in \{0, 6.5, 13\}$, while the phase-augmented analysis showed 0.65, 0.65, and 0.61: the phase-augmented analysis was *less* powerful than the one-component analysis across the entire *pb_strength* range. Two factors produce the inversion. First, at $N = 35$ the rank-deficiency of the full phase-augmented formula required dropping the `Dbc:phase` interaction term, which removes the precise contrast that would have isolated the biomarker-by-treatment-by-phase signal the framework is designed to detect; without that term the phase-augmented analysis pays the cost of additional fixed effects without buying the bias-detection mechanism the framework promises. Second, the predicted bias of the one-component analysis arises in the regression *coefficient*, not in the indicator-level rejection rate. The mean estimated $\widehat{\beta}_{bm:D_{bc}}$ is approximately -2.25 for both analyses across all three *pb_strength* values, so the rejection rate on the biomarker-by-treatment Wald test is not the right summary statistic to detect the framework's predicted bias; the bias should be read off the coefficient mean itself, which the prototype's 100-replicate resolution does not separate from sampling noise at the $N = 35$ scale.

A canonical run at 1{,}000 replicates per cell, larger N (70 and 100), and recovery of the `Dbc:phase` contrast (via a phase-stratified two-stage analysis or a richer trial design that admits the full formula without rank-deficiency) is required before the framework's bias-versus-power trade-offs can be quantified at production rigor. The prototype as configured is a structural check confirming that the analysis machinery runs end-to-end on the three-component DGP, not a substantive test of the manuscript's bias-detection claim. The §6 simulation programme remains the empirical core of this manuscript.

11 Conclusions

Treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Decomposing the response into these components is identifiability-conditional, not optional: the trial design must include phase contrasts that uniquely identify each component, and the analyst must use a methodology that exploits those contrasts. At trial-relevant sample sizes, a phase-augmented linear-mixed analysis captures the practically important part of the decomposition without paying the parametric cost of a full nonlinear-mixed-effects fit. For predictive-biomarker validation, the decomposition is mandatory: a biomarker that predicts the lumped total is not, in general, a pharmacological predictor, and the regulatory and prescribing decisions that motivate the trial cannot be made without separating BR from PB and TV.

12 Conflict of interest

The authors declare no conflicts of interest.

13 Funding

This work received no specific funding.

14 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

15 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/06-decomp-replicates.rds`; Morris ADEMP cell-level summaries at the companion `06-decomp-summary.txt`. Driver scripts are under `analysis/scripts/component-decomposition/`. The pre-registered ADEMP plan is at `analysis/scripts/component-decomposition/00-ademp-pre-registrati` The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/06-component-decomposition/`.

1. Kirsch I, Deacon BJ, Huedo-Medina TB, Scoboria A, Moore TJ, Johnson BT. Initial severity and antidepressant benefits: A meta-analysis of data submitted to the Food and Drug Administration. *PLoS Medicine*. 2008;5(2):e45. doi:10.1371/journal.pmed.0050045

2. Locher C, Kossowsky J, Gaab J, Kirsch I, et al. Moderation of antidepressant and placebo outcomes by baseline severity in late-life depression: A systematic review and meta-analysis. *Journal of Affective Disorders*. 2015;181:50-60. doi:10.1016/j.jad.2015.03.062

3. Sugarman MA, Loree AM, Baltes BB, Grekin ER, Kirsch I. The efficacy of paroxetine and placebo in treating anxiety and depression. *PLoS ONE*. 2014;9(8):e106337. doi:10.1371/journal.pone.0106337

4. Wang RS, Lembo AJ, Kaptchuk TJ, others, Hall KT. Genomic effects associated with response to placebo treatment in a randomized trial of irritable bowel syndrome. *Frontiers in Pain Research*. 2022;2:775386. doi:10.3389/fpain.2021.775386

5. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? An analysis of clinical trials comparing placebo with no treatment. *New England Journal of Medicine*. 2001;344(21):1594-1602. doi:10.1056/NEJM200105243442106

6. Hróbjartsson A, Gøtzsche PC. Is the placebo powerless? Update of a systematic review with 52 new randomized trials. *Journal of Internal Medicine*. 2004;256(2):91-100. doi:10.1111/j.1365-2796.2004.01355.x

7. Hróbjartsson A, Gøtzsche PC. Placebo interventions for all clinical conditions. *Cochrane Database of Systematic Reviews*. 2010;(1):CD003974. doi:10.1002/14651858.CD003974.pub3

990 8. Kent DM, Paulus JK, Klaveren D van, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667

991 9. Kent DM, Klaveren D van, Paulus JK, et al. The PATH statement: Explanation and elaboration. *Annals of Internal Medicine*. 2020;172(1):W1-W25. doi:10.7326/M18-3668

992 10. Kent DM, Steyerberg E, Klaveren D van. Personalized evidence based medicine: Predictive approaches to heterogeneous treatment effects. *BMJ*. 2018;363:k4245. doi:10.1136/bmj.k4245

993 11. Rekkas A, Paulus JK, Raman G, et al. Predictive approaches to heterogeneous treatment effects: A scoping review. *BMC Medical Research Methodology*. 2020;20(1):264. doi:10.1186/s12874-020-01145-1

994 12. Lillie EO, Patay B, Diamant J, Issell B, Topol EJ, Schork NJ. The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*. 2011;8(2):161-173. doi:10.2217/pme.11.7

995 13. Duan N, Kravitz RL, Schmid CH. Single-patient (n-of-1) trials: A pragmatic clinical decision methodology for patient-centered comparative effectiveness research. *Journal of Clinical Epidemiology*. 2013;66(8 Suppl):S21-S28. doi:10.1016/j.jclinepi.2013.04.006

996 14. Gabler NB, Duan N, Vohra S, Kravitz RL. N-of-1 trials in the medical literature: A systematic review. *Medical Care*. 2011;49(8):761-768. doi:10.1097/MLR.0b013e318215d90d

997 15. Schork NJ, Goetz LH. Single-subject studies in translational nutrition research. *Annual Review of Nutrition*. 2017;37:395-422. doi:10.1146/annurev-nutr-071816-064717

998 16. Schork NJ. Randomized clinical trials and personalized medicine: A commentary on Deaton and Cartwright. *Social Science and Medicine*. 2018;210:71-73. doi:10.1016/j.socscimed.2018.04.033

999 17. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15

1000 18. Porcino A, Vohra S. N-of-1 trials, their reporting guidelines, and the advancement of open science principles. *Harvard Data Science Review*. 2022;4(SI3). doi:10.1162/99608f92.a65a257a

1001 19. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004

1002 20. Shamseer L, Sampson M, Bukutu C, Schmid CH, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015: Explanation and elaboration. *Journal of Clinical Epidemiology*. 2015;76:18-46. doi:10.1016/j.jclinepi.2015.05.018

1003 21. Li J, Gao W, Punja S, et al. Reporting quality of N-of-1 trials published between 1985 and 2013: A systematic review. *Journal of Clinical Epidemiology*. 2016;76:57-64. doi:10.1016/j.jclinepi.2015.11.016

1004 22. Liao Z, Qian M, Kronish IM, Cheung YK. Analysis of N-of-1 trials using Bayesian distributed lag model with autocorrelated errors. *Statistics in Medicine*. 2023;42(13):2044-2060. doi:10.1002/sim.9676

1005 23. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137

1006 24. Carrozzo AE, Zimmermann G, Bathke AC, et al. Two-arm crossover randomized controlled trial versus meta-analysis of N-of-1 studies: Comparison of statistical efficiency. *Biometrical Journal*. 2025;67(2):e70045. doi:10.1002/bimj.70045

1007 25. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

1008 26. Kaptchuk TJ, Kelley JM, Conboy LA, et al. Components of placebo effect: Randomised controlled trial in patients with irritable bowel syndrome. *BMJ*. 2008;336(7651):999-1003. doi:10.1136/bmj.39524.439618.25

1009 27. Kaptchuk TJ, Friedlander E, Kelley JM, et al. Placebos without deception: A randomized controlled trial in irritable bowel syndrome. *PLoS ONE*. 2010;5(12):e15591. doi:10.1371/journal.pone.0015591

1010 28. Lembo A, Kelley JM, Nee J, et al. Open-label placebo vs double-blind placebo for irritable bowel syndrome: A randomized clinical trial. *Pain*. 2021;162(9):2428-2435. doi:10.1097/j.pain.0000000000002234

1011 29. Nurko S, Saps M, Kossowsky J, others, Kelley JM. Effect of open-label placebo on children and adolescents with functional abdominal pain or irritable bowel syndrome. *JAMA Pediatrics*. 2022;176(4):349-356. doi:10.1001/jamapediatrics.2021.5750

1012 30. Benedetti F. Placebo effects: From the neurobiological paradigm to translational implications. *Neuron*. 2014;84(3):623-637.

1013 31. Frisaldi E, Shaibani A, Benedetti F. Understanding the mechanisms of placebo and nocebo effects. *Swiss Medical Weekly*. 2020;150:w20340. doi:10.4414/smw.2020.20340

1014 32. Carlino E, Frisaldi E, Benedetti F. Pain and the context. *Nature Reviews Rheumatology*. 2014;10(6):348-355. doi:10.1038/nrrheum.2014.17

1015 33. Carlino E, Piedimonte A, Benedetti F. Nature of the placebo and nocebo effect in relation to functional neurologic disorders. In: *Handbook of Clinical Neurology*. Vol 139.; 2016:597-606. doi:10.1016/B978-0-12-801772-2.00048-5

1016 34. Wager TD, Atlas LY, Lindquist MA, Roy M, Woo CW, Kross E. An fMRI-based neurologic signature of physical pain. *New England Journal of Medicine*. 2013;368(15):1388-1397. doi:10.1056/NEJMoa1204471

1017 35. Raskind MA, Peterson K, Williams T, et al. A trial of prazosin for combat trauma PTSD with nightmares in active-duty soldiers returned from Iraq and Afghanistan. *American Journal of Psychiatry*. 2013;170(9):1003-1010. doi:10.1176/appi.ajp.2013.12081133

1018 36. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.